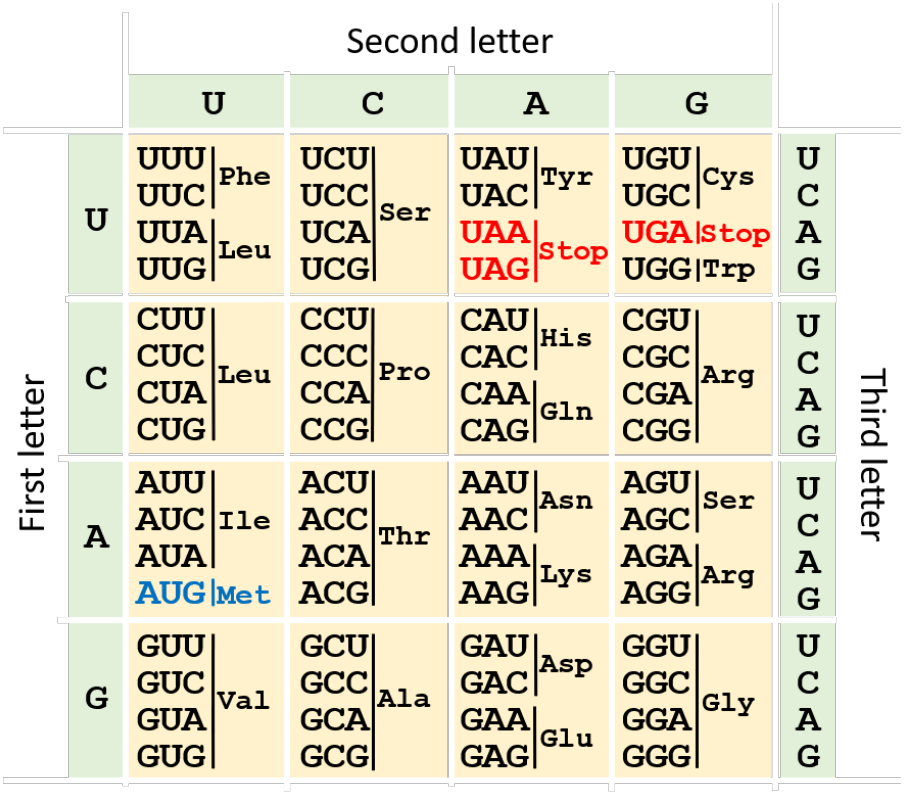
The genetic code

The information contained in the nucleotide sequence of an mRNA molecule is used by the cell for the synthesis of a polypeptide chain. To do this, the cell uses a code known as the **genetic code**.

The genetic code is the set of rules that specifies how nucleotide triplets in mRNA correspond to amino acids or termination signals during translation. Following these rules, an mRNA is "read" to synthesize a polypeptide chain with a specific length and amino acid sequence.

The unit of information in the genetic code is the **codon**. Each codon consists of a triplet of adjacent nucleotides and corresponds to a specific amino acid or stop signal.



From the four nucleotides, there are 64 possible combinations of three-letter nucleotide sequences (codons). Of these 64 codons, 61 code for amino acids, whereas three are stop signals.

If you are curious to know how scientists cracked the genetic code, go to this **link**.

Features of the genetic code

1. The genetic code is nearly **universal**, with some variations in mitochondria and in certain organisms.
2. Codons are read **without overlapping**. Therefore, a single nucleotide cannot be part of two adjacent codons.
3. Within a coding sequence, codons are read consecutively, **without gaps** between them. It is said that there is no punctuation.
4. It is **unambiguous**: each codon specifies a single amino acid or a stop signal.
5. It is **redundant** (also known as **degenerate**): a single amino acid may be encoded by more than one codon (synonymous codons).
6. Redundancy in the genetic code is usually found in the third nucleotide of the triplet.
7. Codons differing by a single nucleotide often encode amino acids with similar chemical properties.
8. **AUG** is the usual i**nitiation codon**, which also codes for methionine. In eukaryotic cytosolic translation, polypeptide synthesis begins with methionine, which may subsequently be removed.
9. There are three **stop codons** (also known as termination or nonsense codons).

Applying the genetic code

Once we know the rules of the genetic code, we can apply them to determine the amino acid sequence encoded by a nucleotide sequence. The reading frame is established by the initiation codon. In the following exercises, we will identify an AUG codon that begins a plausible coding sequence and read successive, nonoverlapping triplets until an in-frame stop codon is reached. The sequences below are written using the DNA alphabet. Replace T with U to obtain the corresponding mRNA sequence before using the codon table.

Let us start with an easy example:

Problem 1: Deduce the amino acid sequence encoded by the short nucleotide sequence indicated below, knowing that the first triplet of nucleotides corresponds to the initiation codon (remember that, by convention, a nucleotide sequence is shown from the 5' to the 3' end).

ATGGAGGACATCTGCATCAACGAGATCAGCATGTACCTGATCTTCGAGTGA

Let us make it harder:

Problem 2: Deduce the amino acid sequence encoded by this nucleotide sequence:

ACCTTCGATCCTCGGGGAGCCCAGGAGACCAGAACATGGACTCCTTCAATTATACCACCC
CTGATTATGGGCACTATGATGACAAGGATACCCTGGACCTCAACACCCCTGTGGATAAAA
CTTCTAACACGCTGCGTGTTCCAGACATCCTGGCCTTGGTCATCTTTGCAGTCGTCTTCC
TGGTGGGAGTGCTGGGCAATGCCCTGGTGGTCTGGGTGACGGCATTCGAGGCCAAGCGGA
CCATCAATGCCATCTGGTTCCTCAACTTGGCGGTAGCCGACTTCCTCTCCTGCCTGGCGC
TGCCCATCTTGTTCACGTCCATTGTACAGCATCACCACTGGCCCTTTGGCGGGGCCGCCT
GCAGCATCCTGCCCTCCCTCATCCTGCTCAACATGTACGCCAGCATCCTGCTCCTGGCCA
CCATCAGCGCCGACCGCTTTCTGCTGGTGTTTAAACCCATCTGGTGCCAGAACTTCCGAG
GGGCTGGCTTGGCCTGGATCGCCTGTGCCGTGGCTTGGGGTTTAGCCCTGCTGCTGACCA
TACCCTCCTTCCTGTACCGGGTGGTCCGGGAGGAGTACTTTCCACCAAAGGTGTTGTGTG
GCGTGGACTACAGCCACGACAAACGGCGGGAGCGAGCCGTGGCCATCGTCCGGCTGGTCC
TGGGCTTCCTGTGGCCTCTACTCACGCTCACGATTTGTTACACTTTCATCCTGCTCCGGA
CGTGGAGCCGCAGGGCCACGCGGTCCACCAAGACACTCAAGGTGGTGGTGGCAGTGGTGG
CCAGTTTCTTTATCTTCTGGTTGCCCTACCAGGTGACGGGGATAATGATGTCCTTCCTGG
AGCCATCGTCACCCACCTTCCTGCTGCTGAAGAAGCTGGACTCCCTGTGTGTCTCCTTTG
CCTACATCAACTGCTGCATCAACCCCATCATCTACGTGGTGGCCGGCCAGGGCTTCCAGG
GCCGACTGCGGAAATCCCTCCCCAGCCTCCTCCGGAACGTGTTGACTGAAGAGTCCGTGG
TTAGGGAGAGCAAGTCATTCACGCGCTCCACAGTGGACACTATGGCCCAGAAGACCCAGG
CAGTGTAGGCGACAGCCTCATGGGCCACTGTGGCCCGATGTCCCCTTCCTTCCCGGCCAT
TCTCCCTCTTGTTTTCACTTCACTTTTCGTGGGATGGTGTTACCTTAGCTAACTAACTCT
CCTCCATGTTGCCTGTCTTTCCCAGACTTGTCCCTCCTTTTCCAGCGGGACTCTTCTCAT
CCTTCCTCATTTGCAAGGTGAACACTTCCTTCTAGGGAGCACCCTCCCACCCCCCACCCC
CCCCACACACACCATCTTTCCATCCCAGGCTTTTGAAAAACAAACAGAAACCCGTGTATC
TGGGATATTTCCATATGGCAATAGGTGTGAACAGGGAACTCAGAATACAGACAAGTAGAA
AGATTCTCGCTTAAAAAAAATGTATTTATTTTATGGCAAGTTGGAAAATATGTAACTGGA
ATCTCAAAAGTTCTTTGGGACAAAACAGAAGTCCATGGAGTTATCTAAGCTCTTGTAAGT
GAGTTAATTTAAAAAAGAAAATTAGGCTGAGAGCAGTGGCTCACGCCTGTAATCCCAGAA
CTTTGGGAGGCTAAGGTGGGTGGATCACCTGAGGTCAAGAGTTCCAGACCAGGCTGGCCA
GCATGGTGAAACCCCGTCTGTACTAAAAATACAAAAAATTAACTGGGCATGGTAGTGGGT
GCCTGTAATCCCAGCTACTTGGGAGGCTGAGGTGGGAGAATTGCTCGAACTTGGAGGTGG
AGGTTGTGGTGAGCCATGATCGCACCACTGCACTCTAGCCTGGGTGACCGAGGGAGGCTC
TGTCTCAAAAGCAAAGCAAAAACAAAAACAAAAACACCTAAAAAACCTGCAGTTTTGTTT
GTACTTTGTTTTTAAATTATGCTTTCTATTTTGAGATCATTGCAAACTCAACACAATTGT
AAGTAATGATACAGAGGGATCTTGTGTACCCTTCACCCAGCCTCCCCCAATGGCAACATC
TTGCAAAACTACAATGTAGTCTCATAACCAGGATATTGACATTGATACAGTGAAGATACA
GGACATTCTCATCACCACAGGGATCCCCAGGATGCCCACTTCCCTCCACCCCCACACCCC
AGCCGTGTCCCTAACCCCTGGCAACCAGGAATCCACTCTCCATTTCTATAATGTTGTCAT
TTCAAGAATGTTATTCAATGGAATCATATAGTATGTAACCTGTTTTGAGCTTAAAAAAAA
AGTATACATGACTTTAATGAGGAAAATAAAAATGAATATTGAAA

In this case, the sequence corresponds to a real mRNA (written using the DNA alphabet), for which the position of the initiation codon and the extent of the coding sequence are unknown. **Do not try to do it by hand.** To solve the problem, we will take advantage of the internet. There are plenty of tools that can help us to determine the amino acid sequence from a nucleotide sequence. For example, we can use this **tool** from the *ExPASy Bioinformatics Resource Portal*. You also have to consider that from any nucleotide sequence, three different reading frames are possible, with many possible start codons. Therefore, when you enter the sequence into the bioinformatics tool, three possible translations are obtained (remember to select only the *forward strand* or you will get six possible solutions). For this exercise, consider the longest **open reading frame** beginning with an AUG codon to be the most likely coding sequence.

Types of mutations

Gene mutations have varying effects on cells, depending on where they occur and whether they alter the function of essential proteins. Knowing the rules of the genetic code allows us to predict, in many cases, the effect that a mutation will have on the encoded protein.

Mutations can be classified by their effect on the DNA sequence or the polypeptide sequence.

Classification of mutations by their effect on the DNA sequence:

- **Substitution** (or **point mutation**): a base is replaced by another.
- **Deletion**: one or more nucleotides are lost.
- **Insertion**: one or more nucleotides are added.
- **Chromosomal rearrangements**: structural changes such as inversions and translocations.

Classification of mutations by their effect on the polypeptide sequence:

- **Missense mutation**: substitution in a codon that changes the amino acid to another amino acid. The impact on protein function will depend on whether the change is "conservative" or "nonconservative" (based on the similarity of the two amino acids), the function of the amino acid in the protein, etc.
- **Synonymous mutation** (or **silent mutation**): substitution that does not change the amino acid sequence. A functional impact cannot be completely ruled out, because the mutation may affect splicing, translation efficiency, or other processes.
- **Nonsense mutation**: a substitution that converts an amino acid-encoding codon into a stop codon, causing a premature termination of translation. It may lead to the production of a truncated protein or to degradation of the mRNA.
- **Frameshift mutation**: results from indels (deletions or insertions) that change the reading frame (that is, the number of affected nucleotides is not a multiple of 3). From that point on, most amino acids are changed, often introducing a premature stop codon.
- **In-frame mutation**: results from indels of 3, or a multiple of 3, nucleotides. The reading frame is preserved. It will result in a protein with amino acids missing or added, but most of the protein remains intact. The functional impact depends on the number and location of the affected amino acids; even a single-amino-acid deletion can severely impair protein function.

Problem 3:

For this in-frame mRNA sequence:

… UCC GGA UAU GUU CAA GCU AGA …

- Deduce the polypeptide sequence
- Deduce the changes in the polypeptide sequence and the type of mutation if:
    - There is a substitution of C by G at position 17:

      … UCC GGA UAU GUU CAA GGU AGA …
    - There is a substitution of U by G at position 18:

      … UCC GGA UAU GUU CAA GCG AGA …
    - There is a substitution of U by A at position 9:

      … UCC GGA UAA GUU CAA GCU AGA …
    - There is an insertion of C before the original nucleotide at position 9:

      … UCC GGA UAC UGU UCA AGC UAG A …
    - There is an insertion of AUC before the original nucleotide at position 10:

      … UCC GGA UAU AUC GUU CAA GCU AGA …

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